validated: restoring the named part removes 93.5–94.0 % of detections (ceiling 93.6–94.1 %)

Photo



the user's photo

Face detection, crop



MediaPipe, 478 landmarks; crop + 35 % per side

Detector



z_f, z_b
4 operations
16×16 map

CLIP ViT-L/14 + LoRA;
decision: $P(\text{fake}) = \sigma(z_f)$

Part mapping



masks: eyes, nose, mouth, skin, contour; name the highest mean evidence; whole face if eyes, nose, mouth all ≥ 0.5

Structured record

```
decision: fake
P: real 0.00, fake 1.00, filter 0.00
ops: eye 0.05, slim 0.01,
     whiten 0.18, smooth 0.15
evidence: mouth 0.99, nose 0.41,
   skin 0.28, contour 0.04, eyes 0.00
named part: mouth
```

every word shown to the user is derived from the record

blind test: VLM region text matched 50.6–59.0 %, chance 50

Fact list (code)

facts, no numbers:
decision is fake; the evidence is concentrated on the mouth

must not say:
real or beautified; any part other than the mouth; any number; certainty or calibration

LLM wording

Qwen2.5-7B-Instruct; no image, no decision, no numbers

> The model's decision is fake. The evidence is concentrated on the mouth.

Code adds numbers

decision score in parentheses, fixed calibration sentence

> The model's decision is fake **(decision score 1.00)**. The evidence is concentrated on the mouth. **The score is the model's raw output, not a calibrated probability, and a high score does not mean the decision is correct.**

Automatic checks

1 decision
2 region
3 operations
4 numbers in record
5 banned phrases
6 length
7 score vs threshold
8 certainty words
9 score on a part
10 no LLM numbers
11 all facts present

11 rules, applied to every text before display (Appendix A)

pass

fail

rewrite once (sampled, checked again)

fail

code template

pass

Output

fake 1.00



decision chip, part on the crop, checked text